

Scaling Microservices with Kubernetes Orchestration

Overview of Cloud-Native Infrastructure

This whitepaper discusses the migration of monolithic architectures to cloud-native microservices. We explore container orchestration systems and high availability resource scaling.

Kubernetes Container Orchestration

We use Kubernetes to manage dockerized application nodes. The microservices are packaged in Docker containers and deployed across multi-tenant cluster nodes. Auto-scaling rules scale compute resources dynamically based on CPU throughput.

Load Balancing and Traffic Gateways

To manage external traffic, we configure distributed load balancers and secure proxy gateways. This ensures low latency routing between serverless lambda functions and containerized web engines.

Summary of Cloud Benefits

Migrating to cloud environments like AWS and Microsoft Azure reduces infrastructure cost. Integrating DevOps pipelines and Infrastructure as Code (IaC) ensures repeatable deployments and minimal operational overhead.